

Multi-Salvo Ballistic Missile Defense Using Virtual Targets

Jordan A. Graham* and David R. Jacques†

Air Force Institute of Technology, Wright-Patterson AFB, OH, 45433, USA

Isaac E. Weintraub‡

Air Force Research Lab, Wright-Patterson AFB, OH, 45433, USA

Edward D. White§

Air Force Institute of Technology, Wright-Patterson AFB, OH, 45433, USA

The proliferation of advanced ballistic threats necessitates adaptive defensive strategies. This paper builds upon previous research in a 2-D environment by investigating the performance of multi-salvo doctrines in a three-dimensional simulation featuring ballistic trajectories [1]. We evaluate three strategies: a single-salvo baseline, a sequential shoot-look-shoot engagement, and a proactive multi-salvo strategy employing Virtual Targets (VTs). Optimal Weapon-Target Assignments are determined using a multi-objective optimization model, implemented in Julia, which balances kinematic intercept costs with a time-dependent Probability of Kill (Pk) model. A full factorial Monte Carlo simulation was conducted to analyze the sensitivity of mission outcomes to force ratios, kinematic advantages, and weapon lethality. Results confirm that multi-salvo strategies remain significantly superior to a single salvo in reducing the number of surviving attackers ("leakers"). The use of VTs consistently increases standoff distance, enhancing the defensive posture. These findings provide quantitative insights to inform the development of next-generation battle management systems for countering complex threats.

Nomenclature

α	=	Multi-objective weighting factor, $\alpha \in [0, 1]$
β	=	Cost function scaling factor
C	=	Global multi-objective cost function value
$P_{k,0}$	=	Maximum kill probability at zero intercept time
$P_{k,j}(t)$	=	Probability of kill for a single defender

*Masters Student, AFIT/ENV, jordan.graham.9@us.af.mil.

†Professor, AFIT/ENV, david.jacques@us.af.mil, Associate fellow, AIAA

‡Senior Engineer, Control Science Center, Air Force Research Laboratory, Lifetime Associate Fellow, AIAA.

§Professor, AFIT/ENC, Stat.Associates@gmail.com, and AIAA Member Grade (if any) for fourth author (etc.).

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$P_k(t)$	=	Probability of kill to neutralize an attacker in a single engagement
P_s	=	Probability of survival for an attacker in a single engagement
σ	=	Pk Decay Constant
T	=	Defender total effort in seconds
t_j	=	The j th defender's time of flight from launch to intercept in seconds
θ	=	Defender heading change angle at a VT in radians
w	=	Weighting constant for the maneuver penalty at a VT

I. Introduction

THE proliferation of advanced aerial threats presents a formidable challenge to national and allied security. Recent global conflicts have highlighted the necessity of robust, multi-layered defensive systems capable of countering large-scale, coordinated attacks. A foundational concept in designing such systems is the Weapon-Target Assignment (WTA) problem, which seeks the optimal allocation of defensive assets against a set of incoming threats [2, 3]. While traditional doctrine often favors a single, large salvo of interceptors (Static Weapon Target Assignment) [3–5], this approach is resource-intensive and lacks the adaptability required for evolving threat scenarios.

An alternative and more dynamic strategy is the use of multi-salvo engagement, a layered defense in which interceptors are deployed in sequential waves [4–7]. This doctrine provides multiple opportunities to engage threats and allows for adaptive decision-making between salvos [8]. Preliminary studies investigated the effectiveness of multi-salvo strategies within a simplified two-dimensional, constant-velocity framework [1, 9, 10]. That work compared a baseline missile defense scenario using a single-salvo against a shoot-look-shoot strategy and a multi-salvo strategy employing Virtual Targets (VTs)—calculated intermediate waypoints used to pre-position assets. The findings indicated that both multi-salvo doctrines significantly reduced the number of surviving attackers ("leakers") and that the VT-based approach offered the dual benefit of increasing standoff distance while reducing engagement time [1].

The simplified 2-D model served as a foundational step. This paper expands on that work to test VT strategies against different threat profiles, specifically ballistic trajectories. Similarly to the aforementioned work, this paper presents a stochastic simulation framework. The three missile defense strategies (single-salvo, shoot-look-shoot, and multi-salvo using VTs) are re-evaluated against attackers following ballistic paths with defenders also following ballistic paths. Weapon target assignments are optimized using a multi-objective function that balances the kinematic "energy cost" to intercept with a time-dependent Probability of Kill (Pk) model.

The insights derived from this higher fidelity model and analysis aim to provide a deeper understanding of the trade-offs inherent in modern layered defense and inform the development of next-generation battle management systems.

II. Materials and Methods

The engagement model presented in this work extends the stochastic framework established in [1] by transitioning from 2-D planar movement to a higher fidelity 3-D battle-space. This section details the physical environment, the 3-D engagement doctrines, and the mathematical formulation of the optimization model.

A. Environment and Solver

The simulation environment is defined in a three-dimensional Cartesian frame (x, y, z) under a flat-earth assumption. Unlike the constant-velocity assumptions utilized in preliminary studies, both attackers and defenders are modeled as ballistic objects subject to a uniform gravitational field. Atmospheric drag and Earth curvature are neglected to isolate the kinematic trade-offs of the engagement strategies. Unlike the Apollonius circle or Earliest Intercept Geometry (EIG) for 2-D solutions [1, 10, 11], the 3-D ballistic intercept point is determined numerically. For every potential defender-attacker pair, a non-linear system of equations is solved using a Newton-Raphson approach (NLSolve) to find the launch azimuth, elevation, and time-of-flight required for convergence.

B. Engagement Architectures

Three defensive doctrines are evaluated to determine their effectiveness against ballistic threat profiles.

1. Case 1: Single Salvo Engagement

Case 1 represents a traditional defensive posture. A coordinated salvo of defenders is launched from the baseline ($x = 1000, z = 0$) to intercept inbound threats, and the defenders are spaced evenly in the y -direction at 6 different launch sites ($y = -2500, -1500, -500, 500, 1500, \text{ and } 2500$). Attackers are initialized at a fixed distance of $x = 100000$ and $z = 0$ with randomized launch positions (y), targeting the protected asset at the origin. This serves as the kinematic baseline for leaker-rate comparisons. Figure 1 depicts an example of this engagement. The red markers represent attacker starting locations projecting to travel along the red dashed line toward a common target. The green markers represent the defenders projecting to follow the green paths to intercept the attackers before they reach the defended asset.

2. Case 2: Sequential Salvo Engagement (Shoot-Look-Shoot)

This doctrine evaluates the benefit of temporal layering. The total defensive force is divided equally into two waves; the first wave engages the primary threat, followed by an instantaneous battle damage assessment (BDA). Any surviving "leakers" are re-engaged by a second wave launched from the baseline. This case provides a benchmark for adaptive responses according to real-time feedback. Figure 2 displays the shoot-look-shoot scenario. The attackers are still red; however, the defenders are split in two salvos of 6 defenders each. The green paths represent the first salvo, while the blue represents the 2nd salvo tracking toward the leakers.

3. Cases 3 and 4: Asynchronous Salvo with Virtual Targets (VTs)

Cases 3 and 4 explore the use of VTs. At $t = 0$, the optimizer selects a set of optimal coordinates (x, y, z) within a predefined bounding box to serve as VTs. In Case 3, the maximum allowable number of VTs to select is low, whereas Case 4 uses a higher maximum allowable number of VTs. After the first salvo is launched directly toward the incoming attackers, the second salvo is launched asynchronously toward the VTs. Their launch times are set so they each arrive at their VTs at the moment of the BDA. Upon arrival at the VT, the defenders are assigned to any remaining leakers and their flight paths are adjusted accordingly. This doctrine seeks to maximize the defended footprint by pre-positioning assets downrange. In Figure 3, the dashed yellow field represents the bounding box for all the candidate VTs [10], and the selected VTs are the cyan diamonds. The dashed gray and blue lines portray the second salvo of defenders.

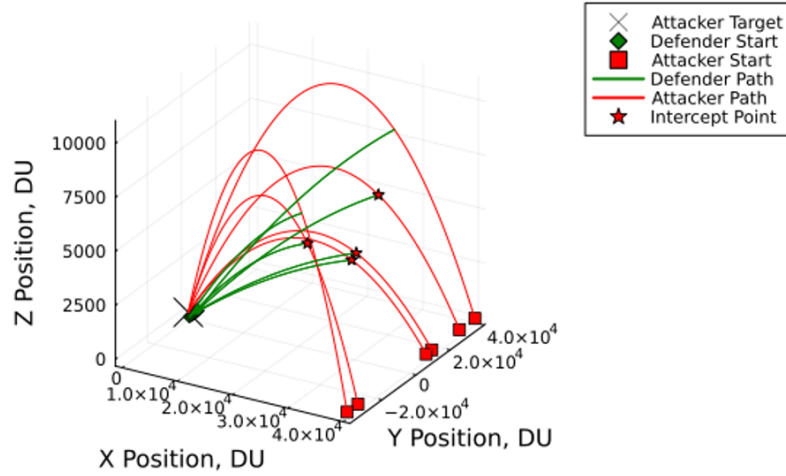


Fig. 1 Case 1: Single Salvo Engagement

C. Objective Function and Cost Metrics

The optimization problem is formulated as a Mixed-Integer Non-Linear Program (MINLP). The solver minimizes a global cost function C which balances kinematic efficiency and mission success. Specifically, it aims to minimize defensive effort (D) and the expected survival probability of the attackers (P_s):

$$C = \alpha T + (1 - \alpha)\beta P_s \quad (1)$$

Total effort, T , represents the cumulative flight time and associated energy penalty,

$$T = t_j + w\theta \quad (2)$$

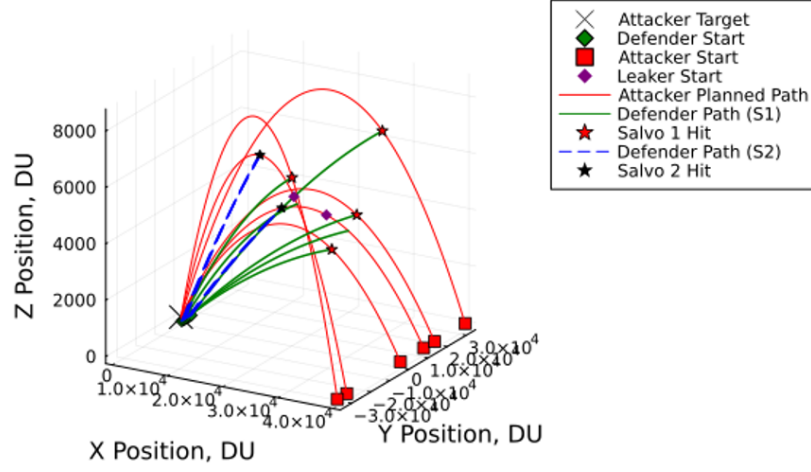


Fig. 2 Case 2: Sequential Salvo Engagement

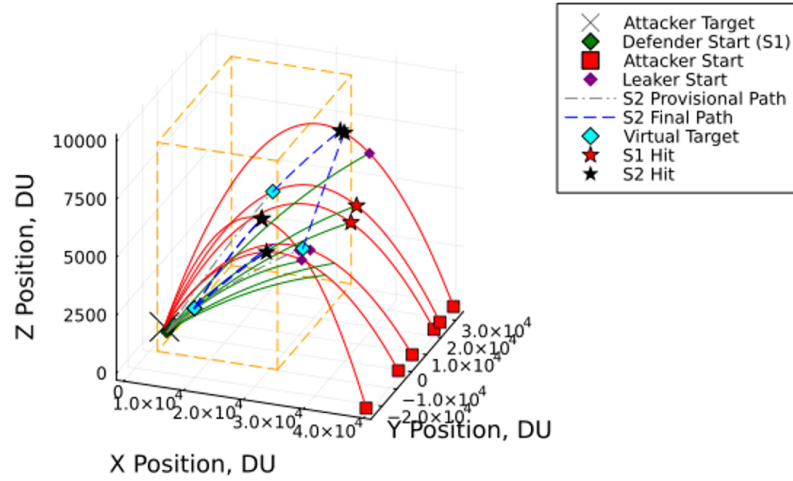


Fig. 3 Cases 3 & 4: Asynchronous Salvo with Virtual Targets

where t_j is the j th defender's time of flight from launch to intercept, and $w\theta$ is maneuver penalty to simulate energy loss from velocity vector redirection at the VT. This ensures that the optimizer prioritizes trajectories that avoid aggressive maneuvers, reserving kinetic energy for terminal engagement.

The probabilistic component is governed by a time-dependent Probability of Kill (Pk) model for a single defender, adapted from Merkulov [12]:

$$P_{k,j}(t) = P_{k_0} e^{-\frac{t_j^2}{2\sigma^2}} \quad (3)$$

where P_{k_0} is the maximum probability of kill at $t_j = 0$, and σ is a decay constant that dictates how rapidly the Pk diminishes with time. A larger σ corresponds to a more capable defender whose Pk degrades more slowly.

When multiple defenders are assigned to a single attacker, the probability of kill is calculated as the product of the individual P_k 's, assuming independent engagement outcomes:

$$P_k(t) = 1 - \prod_{j=1}^n (1 - P_{k,j}(t_j)) \quad (4)$$

where n is the total number of defenders assigned to a single attacker, and $P_{k,j}(t_j)$ represents the probability of kill for the j th defender at time t_j .

The optimization objective is to minimize the expected number of surviving attackers, and the probability of an attacker surviving an engagement, P_s , is defined as:

$$P_s = 1 - P_k(t) \quad (5)$$

D. Method

A Monte Carlo analysis was performed using a full factorial design. The search space for VTs was expanded to a 3-D bounding box, allowing the optimizer to leverage altitude as a variable. For sake of the Design of Experiments, Case 3 was split into Case 3 and 4 to address the number of VTs chosen. Case 3 represents a scenario where there are a low maximum allowable number of VTs to select, while Case 4 has a high maximum allowable number of VTs. In both virtual target cases, multiple defenders will share VTs, so it is assumed that defenders co-located at the same VT are separated by altitude. For all 4 cases, three factors were investigated: defender-to-attacker (Def-Atk) force ratio, defender-to-attacker speed ratio, and the P_k decay rate σ . Each factor was tested at two levels (high and low) resulting in 8 unique scenarios. The simulation was executed with 30 replications for each scenario, resulting in 240 runs per case. Seven measures of effectiveness were tracked with a primary focus on the reduction of leaker rates, and the increase in the defended stand-off range. The DOE matrix is shown in Table 2.

1. Primary Response Variables:

- 1) **Total Leakers:** The average number of attackers that are not successfully intercepted by any salvo.
- 2) **Stand-off Distance:** The average x-coordinate of the successful intercept closest to the asset boundary line at $x=0$.

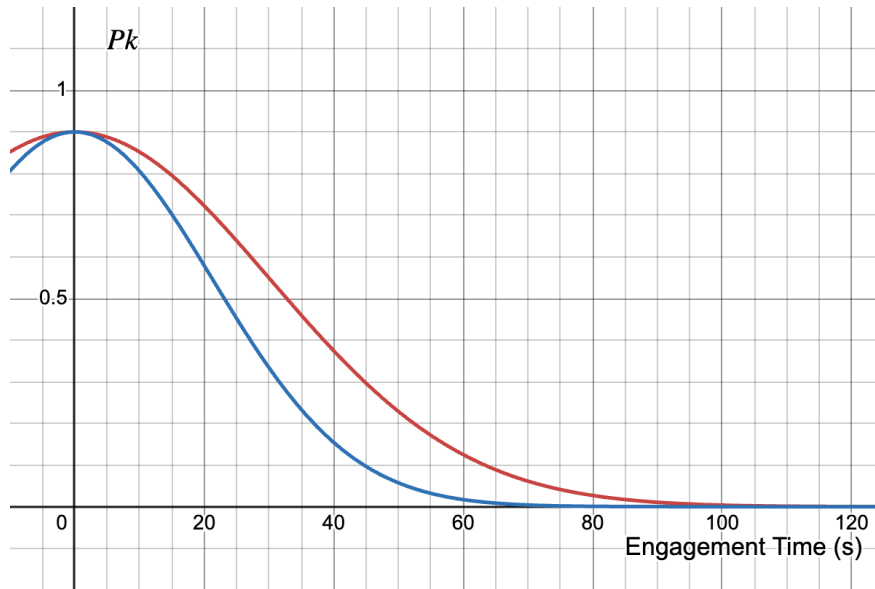
2. Input Factors:

- 1) **Defender-to-Attacker Ratio:** This ratio represents the numerical advantage of the defense. A fixed force of 12 defenders was used across all cases, while the number of attackers was varied. The low setting was a 2:1 ratio (6 attackers), and the high setting was a 3:1 ratio (4 attackers).

Table 2 DOE Matrix

Def-Atk Ratio	Speed Ratio	σ
2	1.3	21.3
2	1.3	30.2
2	2	21.3
2	2	30.2
3	1.3	21.3
3	1.3	30.2
3	2	21.3
3	2	30.2

- 2) **Speed Ratio:** This ratio defines the kinematic advantage of the defenders. Defender speed was fixed at an initial velocity of 1.5 DU/s. The low setting was a ratio of 1.3 (attacker initial velocity = 1.15 DU/s), and the high setting was 2.0 (attacker initial velocity = 0.75 DU/s).
- 3) **Pk Decay σ :** The decay constant σ was selected to produce specific warhead lethality (Pk) values for a benchmark engagement scenario to reflect the condition where an interceptor may lose its kinematic advantage over time. The longest recorded engagement in this scenario is 27.17 sec, with a slow attacker intercepted at the furthest distance from the target. This time was used to set low setting of $\sigma = 21.3$ resulting in a Pk of 0.4 and the high setting of $\sigma = 30.2$ yielding a Pk of 0.6, representing a more advanced weapon system. The Pk Decay rate versus time is depicted in Figure 4. The decay of Pk with time is intended to account for the loss of kinetic energy associated with a boost-coast thrust profile for a defender missile.

**Fig. 4 Pk Decay: Blue ($\sigma = 21.3$); Red ($\sigma = 30.2$)**

III. Results

The Monte Carlo simulation results characterize the performance of the proposed engagement doctrines for a 3-D ballistic environment. The following analysis evaluates the stochastic convergence of leaker rates across the four cases and provides a sensitivity analysis of the parameters governing the Virtual Target strategy.

A. Case Comparison

Consistent with the 2-D planar baseline, multi-salvo operations (Cases 2-4) demonstrate a statistically significant reduction in residual threat leakage compared to the single-salvo baseline (Case 1). As detailed in Figure 5, Case 1 resulted in an average of 0.48 leakers in each run, whereas the multi-salvo cases reduced this value to approximately 0.10. There was a small difference in leaker rates between a pure shoot-look-shoot strategy and the virtual target cases; however, the difference between these means is not statistically significant. This was confirmed by an Odds Ratio analysis which also revealed that in the tested scenarios, single salvo strategies are more than 9 times more likely to have a leaker than multi-salvo strategies.

Table 3 Odds Ratio Analysis for Leakers by Case

Level 1	Level 2	Odds Ratio	P-Value
1	2	9.75	<0.0001
1	3	9.01	<0.0001
1	4	9.01	<0.0001

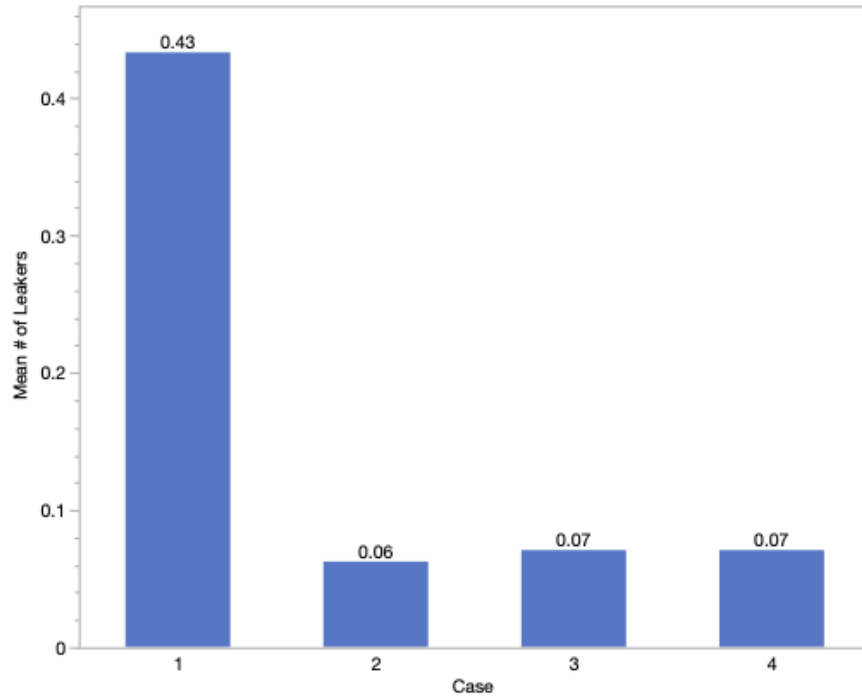


Fig. 5 Average Leakers per Case

The employment of VTs results in a substantial expansion of the defended footprint. The median stand-off distance increased from 16 DU in Case 2 to around 19 DU in Cases 3 and 4 — an 18.75% increase in stand-off distance (see Fig 6). While not considered in this research, the increased stand-off distance may allow for an additional salvo to catch any leakers.

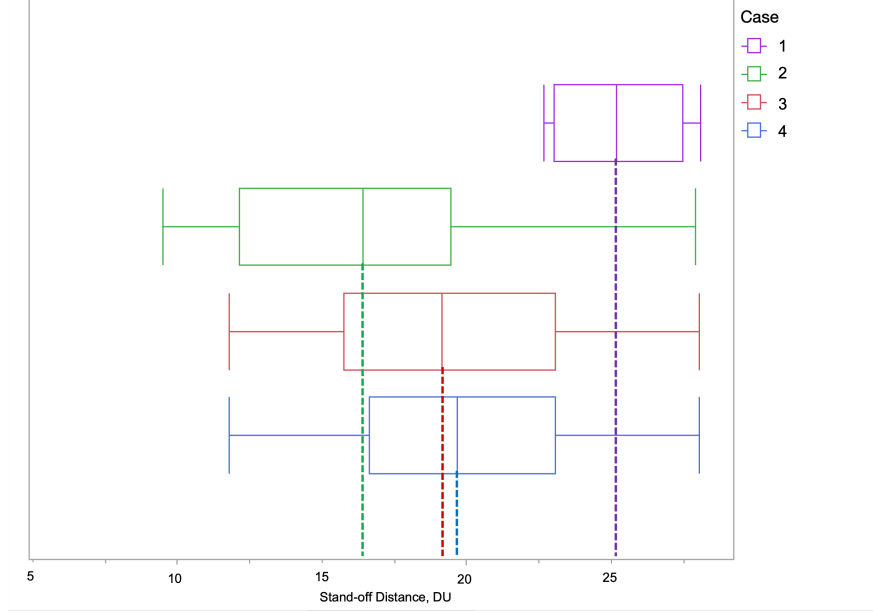


Fig. 6 Stand-off Distance

B. Sensitivity Analysis for VT Cases

A multiple regression analysis was performed on the aggregated results of Cases 3 and 4 to quantify the sensitivity of mission outcomes to the three primary input factors and their interactions. The key findings for each Measure of Effectiveness (MOE) are detailed below.

1. Factors Affecting Total Leakers

As detailed in Tables 4 and 5, the probability of threat leakage is most sensitive to the defender-to-attacker ratio; however, the Pk decay (σ) and speed ratio are still significant. This result is expected, as numerical superiority is the primary driver of success. Furthermore, an interaction exists between the defender-to-attacker ratio and Pk decay (σ). As the number of attackers decrease, the Pk decay rate has a diminishing effect on preventing the number of leakers. In this simulation, numerical superiority appears to bring a greater advantage than an increased probability of kill.

This changes slightly when assessing the factors that affect whether or not there are any leakers at all. While it appears all factors affect the total number of leakers per scenario, the defender-to-attacker ratio and speed-ratio determine if any leaker gets through the defenders.

Table 4 Statistically Significant Factors Affecting Total Leakers for Cases 3 and 4

Factor	P-Value
Def-Atk Ratio	0.0061
Pk Decay Constant (σ)	0.0200
Speed Ratio	0.0200
Def-Atk Ratio & Pk Decay Constant (σ)	0.0200

Table 5 Statistically Significant Factors Affecting Any Leaker for Cases 3 and 4

Factor	P-Value
Def-Atk Ratio	0.0331
Speed Ratio	0.0187

2. Statistical Significance of Factors Affecting Stand-off Distance

The analysis of Stand-off Distance (Table 6) supports analysis from the 2-D kinematic models [1]. Speed Ratio and Def-Atk Ratio are the primary drivers of stand-off distance. However, in the 3-D ballistic environment, these are the only two significant factors, whereas the previous study found the Pk Decay (σ) to be significant as well.

Table 6 Statistically Significant Factors Affecting Stand-off Distance for Case 3

Factor	P-Value
Speed Ratio	<0.0001
Def-Atk Ratio	<0.0001

Table 7 Statistically Significant Factors Affecting Stand-off Distance for Case 4

Factor	P-Value
Speed Ratio	<0.0001
Def-Atk Ratio	<0.0001

The parameter analysis revealed that in these tested scenarios, a higher speed ratio increased the stand-off distance by about 3.7 DU while an increase from the low to high Def-Atk ratio increased the stand-off distance by about 2.7 DU.

IV. Conclusion

The transition from 2-D kinematic modeling to a 3-D ballistic environment supports the findings from the previous studies and reinforces the fundamental advantage using virtual targets. Furthermore, the employment of VTs offers a strategic benefit with a negligible cost to mission success. The forward advanced defensive positioning utilized in Cases 3 and 4 increased the engagement stand-off distance by 3 DU—representing an 18.75% expansion of the defended footprint—while the associated increase in leaker rates remained statistically insignificant. This trade-off is highly favorable for modern air defense architectures. Any increase in stand-off distance may provide a Battle Management

System (BMS) with more time to allow for additional salvos that would otherwise be impossible under a traditional missile defense doctrine. Furthermore, increased distance from interceptions also may provide additional protection from missile debris.

This research was conducted on a Macbook Pro with an Apple M3 Pro chip and 36 GB of RAM. Among the tested cases, 2 different optimizers were used: one for direct engagements, and the other for finding the VTs. In the virtual target case, the direct engagement optimizer was used twice: once for the first salvo, and the second for target reassignment after the second salvo defenders arrived at the VTs. Across all trials, as the optimizer calculated assignments for both the first salvo and the reassignment of the second salvo to engage the leakers, 99.5% of these optimizations had a computation time under 0.21 sec. However, the computation time to optimize the virtual target selection had more variation. The distribution is skewed right with a median time of 0.6 sec and a mean of 1.3 sec. However, some scenarios took significantly longer to optimize with a maximum of 23.25 sec. This is likely due to different attacker starting locations and scenario configurations. In reality, missile defense systems will ideally coordinate WTA within milliseconds, so future work on multi-salvo virtual target strategies should focus on heuristics or alternative solutions to manage the computational effort efficiently.

Ultimately, these findings suggest that the strategic value of virtual targets lies in their ability to find time and space in high-velocity engagements. The exact solution is numerically costly and there can be a significant variation in the time to compute; however, the utility of virtual targets is promising and may provide an advantage in multi-salvo missile defense operations.

A. Future Work

The framework developed in this research provides a foundation for numerous avenues of future investigation, beginning with the optimization of VT search spaces. Future studies should investigate the sensitivity of the VT doctrine to the geometry and density of the search volume, specifically examining the utility of increasing the maximum allowable number of VTs and determining the optimal positioning and characterization of the VT grid space. To enhance realism, high-fidelity environmental modeling should incorporate atmospheric drag and Earth curvature to evaluate the impact of aerodynamic degradation on interceptor reachability and VT effectiveness. While this research calculated exact intercept times and locations, operational defenders will likely rely on guidance laws. Future work should evaluate how the integration of guidance laws affect both computational demand and performance. Furthermore, the scope of this research could be expanded to incorporate adversarial maneuverability and heterogeneous assets. This may include evading attackers reacting to incoming defenders or a mix of defender and attacker types, with varying speeds, weapon capabilities (Pk models), and roles. Finally, future research in this area should aim to apply heuristics and approximation methods to decrease the compute time for VT selection while achieving comparable performance. By exploring these areas, a more comprehensive understanding of the trade-offs and sensitivities inherent in modern air and missile defense

can be achieved.

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